

Homework Assignment 1: Slab-Geometry Boltzmann Equation

Tal Ramot

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Question 1 - Useful identity for the solution of the time-dependent slab-geometry Boltzmann equation

In this question we are asked to prove the following identity:

$$\psi(x, \mu, t; c) = ce^{-(1-c)v\Sigma_t t} \psi(cx, \mu, ct; 1) \quad (1)$$

We will do it in two parts. First, we will write down the equation that $\tilde{\psi} = \psi(cx, \mu, ct; 1)$ satisfies, and then we will use it in order to prove that the product $ce^{-(1-c)v\Sigma_t t} \tilde{\psi}$ satisfies the same equation as $\psi(x, \mu, t; c)$.

The equation for $\tilde{\psi}$

Notation: We will denote by $\Psi(X, \mu, T) \equiv \psi(X, \mu, T; 1)$ the $c = 1$ solution written in its own variables, and by Ψ_T and Ψ_X its partial derivatives with respect to them, so that $\tilde{\psi}(x, \mu, t) = \Psi(cx, \mu, ct)$ with $X = cx$ and $T = ct$. **This notation is used to avoid confusion when using the chain rule, as will be done immediately.** First, Ψ satisfies the $c = 1$ equation in the variables X and T :

$$\frac{1}{v} \Psi_T + \mu \Psi_X + \Sigma_t \Psi = \frac{\Sigma_t}{2} \int_{-1}^1 \Psi d\mu' \quad (2)$$

which after applying the chain rule, $\Psi_T = \frac{1}{c} \frac{\partial \tilde{\psi}}{\partial t}$ and $\Psi_X = \frac{1}{c} \frac{\partial \tilde{\psi}}{\partial x}$, and multiplying by c , becomes:

$$\frac{1}{v} \frac{\partial \tilde{\psi}}{\partial t} + \mu \frac{\partial \tilde{\psi}}{\partial x} + c\Sigma_t \tilde{\psi} = \frac{c\Sigma_t}{2} \int_{-1}^1 \tilde{\psi} d\mu' \quad (3)$$

So $\tilde{\psi}$ obeys the same equation as $\psi(x, \mu, t; c)$ except that it is removed at the rate $c\Sigma_t$ instead of Σ_t .

Proving the identity

In this part we are going to simply substitute the RHS of 1 into the LHS of the equation for $\psi(x, \mu, t; c)$ and show that it satisfies the same equation. After substitution and application of the chain rule we

get:

$$\text{LHS} = \frac{1}{v} \left[c(c-1)v\Sigma_t e^{-(1-c)v\Sigma_t t} \tilde{\psi} + c e^{-(1-c)v\Sigma_t t} \frac{\partial \tilde{\psi}}{\partial t} \right] + \mu c e^{-(1-c)v\Sigma_t t} \frac{\partial \tilde{\psi}}{\partial x} + \Sigma_t c e^{-(1-c)v\Sigma_t t} \tilde{\psi} \quad (4)$$

$$= c e^{-(1-c)v\Sigma_t t} \left[\frac{1}{v} \frac{\partial \tilde{\psi}}{\partial t} + \mu \frac{\partial \tilde{\psi}}{\partial x} + \Sigma_t \tilde{\psi} + (c-1)\Sigma_t \tilde{\psi} \right] \quad (5)$$

$$= c e^{-(1-c)v\Sigma_t t} \left[\frac{1}{v} \frac{\partial \tilde{\psi}}{\partial t} + \mu \frac{\partial \tilde{\psi}}{\partial x} + c\Sigma_t \tilde{\psi} \right] \quad (6)$$

$$= c e^{-(1-c)v\Sigma_t t} \frac{c\Sigma_t}{2} \int_{-1}^1 \tilde{\psi} d\mu' = \frac{c\Sigma_t}{2} \int_{-1}^1 c e^{-(1-c)v\Sigma_t t} \tilde{\psi} d\mu' \quad (7)$$

where 3 was used in the last line, and this is exactly the RHS of the equation for $\psi(x, \mu, t; c)$.

Question 2 - Derivation of the time-dependent planar scalar flux for a plane source

The relation between a plane source and a point source as studied in class is given by:

$$\phi_{\text{pl}}(x) = 2\pi \int_{|x|}^{\infty} \phi_{\text{pt}}(r) r dr \quad (8)$$

The solution taken from J. C. J. Paasschens is the point-source scalar flux $\phi_{\text{pt}}(r, t)$ for $c = 1$, and therefore the planar-source scalar flux is given by the integration given above.

$$\phi_{\text{pl}}(x, t) = 2\pi \int_{|x|}^{\infty} \frac{e^{-v\Sigma_t t}}{4\pi r^2} \delta(r - vt) r dr + 2\pi \int_{|x|}^{\infty} \frac{(1 - r^2/v^2 t^2)^{1/8}}{(4\pi vt/(3\Sigma_t))^{3/2}} e^{-v\Sigma_t t} G\left(v\Sigma_t t \left[1 - \frac{r^2}{v^2 t^2}\right]^{3/4}\right) \Theta(vt - r) r dr \quad (9)$$

$$= \frac{e^{-v\Sigma_t t}}{2vt} \Theta(vt - |x|) + \Theta(vt - |x|) \frac{2\pi e^{-v\Sigma_t t}}{(4\pi vt/(3\Sigma_t))^{3/2}} \int_{|x|}^{vt} \left(1 - \frac{r^2}{v^2 t^2}\right)^{1/8} G\left(v\Sigma_t t \left[1 - \frac{r^2}{v^2 t^2}\right]^{3/4}\right) r dr \quad (10)$$

$$\stackrel{\xi=r/vt}{=} \frac{e^{-v\Sigma_t t}}{2vt} \Theta(vt - |x|) + \Theta(vt - |x|) \frac{2\pi e^{-v\Sigma_t t} (vt)^{1/2}}{(4\pi/(3\Sigma_t))^{3/2}} \int_{\frac{|x|}{vt}}^1 (1 - \xi^2)^{1/8} G\left(a [1 - \xi^2]^{3/4}\right) \xi d\xi \quad (11)$$

$$\stackrel{u=1-\xi^2}{\stackrel{w=au^{3/4}}{=}} \frac{e^{-v\Sigma_t t}}{2vt} \left[1 + \sqrt{\frac{3}{\pi}} \int_0^{w_0} \sqrt{w} G(w) dw \right] \Theta(vt - |x|) \quad (12)$$

with $w_0 = a \left(1 - \frac{x^2}{(vt)^2}\right)^{3/4}$, the powers combining as $\frac{1}{6} + \frac{1}{3} = \frac{1}{2}$ and the Σ_t dependence cancelling in the last step (I used the assistance of AI in this algebraic derivation). Finally, with the interpolation formula of Paasschens, $G(w) = e^w \sqrt{1 + b/w}$ with $b = 2.026$, the integral has a closed form:

$$\int_0^{w_0} e^w \sqrt{w + b} dw = e^{w_0} \sqrt{w_0 + b} - \sqrt{b} - \frac{\sqrt{\pi}}{2} e^{-b} \left[\text{erfi}\left(\sqrt{w_0 + b}\right) - \text{erfi}\left(\sqrt{b}\right) \right] \quad (13)$$

which **closes the derivation of the planar-source scalar flux for $c = 1$** . Of course the relation proved in Q1 can be used to get the solution for $c \neq 1$ using a simple scaling and multiplication by an exponential factor of the solution for $c = 1$.

Question 3 - Classical and asymptotic diffusion against the exact solution

Derivation of the time-dependent diffusion Green's function

The time-dependent diffusion equation is:

$$\frac{1}{v} \frac{\partial \phi}{\partial t} - D \frac{\partial^2 \phi}{\partial x^2} + \Sigma_a \phi = S(x, t) \quad (14)$$

Take $S = \delta(x)\delta(t)$ and multiply (14) by v :

$$\frac{\partial \phi}{\partial t} - Dv \frac{\partial^2 \phi}{\partial x^2} + v\Sigma_a \phi = v\delta(x)\delta(t) \quad (15)$$

Step 1: the pulse source is an initial condition. Integrate (15) over $t \in (0^-, 0^+)$. The diffusion and absorption terms are bounded and contribute nothing in the limit, so only the time derivative survives:

$$\phi(x, 0^+) - \phi(x, 0^-) = v\delta(x) \quad \implies \quad \phi(x, 0^+) = v\delta(x) \quad (16)$$

since nothing precedes the pulse. For $t > 0$ the equation is source-free, so the problem is now a pure initial-value problem (this is the same trick as in Assignment 1, Question 2)

Step 2: absorption factors out exactly. As performed in the course, substituting $\phi(x, t) = e^{-v\Sigma_a t} u(x, t)$ into (15) for $t > 0$, the $-v\Sigma_a u$ produced by the time derivative cancels the absorption term identically and leaves the pure heat equation

$$\frac{\partial u}{\partial t} = Dv \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0^+) = v\delta(x) \quad (17)$$

Note this step is untouched by the sign of $1 - c$: for $c > 1$ the exponential grows instead of decays, and nothing else changes.

Step 3: Fourier transform. With $\hat{u}(k, t) = \int_{-\infty}^{\infty} u(x, t) e^{-ikx} dx$, the spatial derivative becomes multiplication by $-k^2$ and the PDE becomes an ordinary differential equation in t for each k separately:

$$\frac{d\hat{u}}{dt} = -Dvk^2 \hat{u}, \quad \hat{u}(k, 0^+) = v \quad \implies \quad \hat{u}(k, t) = v e^{-Dvk^2 t} \quad (18)$$

the initial value being v because the transform of $v\delta(x)$ is the constant v — a delta excites every wavenumber equally.

Step 4: invert. Completing the square in the exponent, $-Dvt k^2 + ikx = -Dvt \left(k - \frac{ix}{2Dvt}\right)^2 - \frac{x^2}{4Dvt}$, and using $\int_{-\infty}^{\infty} e^{-\alpha k^2} dk = \sqrt{\pi/\alpha}$:

$$u(x, t) = \frac{v}{2\pi} \int_{-\infty}^{\infty} e^{-Dvk^2 t} e^{ikx} dk = \frac{v}{\sqrt{4\pi Dvt}} e^{-x^2/(4Dvt)} \quad (19)$$

Restoring the factor from Step 2 gives the Green's function:

$$\boxed{\phi_{\text{diff}}(x, t; c) = \frac{v e^{-(1-c)\Sigma_a t}}{\sqrt{4\pi Dvt}} \exp\left(-\frac{x^2}{4Dvt}\right)} \quad (20)$$

which in the units of this assignment, $\Sigma_t = v = 1$ and $D = D_0(c)$, is the form implemented in `src/homework2/diffusion.py`:

$$\phi_{\text{diff}}(x, t; c) = \frac{e^{-(1-c)t}}{\sqrt{4\pi D_0(c)t}} e^{-x^2/(4D_0(c)t)} \quad (21)$$

A second route: Fourier in space, then Laplace in time

Define

$$\hat{\phi}(k, t) = \int_{-\infty}^{\infty} \phi(x, t) e^{-ikx} dx, \quad \tilde{\phi}(k, s) = \int_{0^-}^{\infty} \hat{\phi}(k, t) e^{-st} dt \quad (22)$$

and apply both to (15) term by term, with $\phi(x, 0^-) = 0$:

$$\frac{\partial \phi}{\partial t} \longrightarrow s \tilde{\phi} - \hat{\phi}(k, 0^-) = s \tilde{\phi} \quad (23)$$

$$-Dv \frac{\partial^2 \phi}{\partial x^2} \longrightarrow Dvk^2 \tilde{\phi} \quad (24)$$

$$v\Sigma_a \phi \longrightarrow v\Sigma_a \tilde{\phi} \quad (25)$$

$$v \delta(x) \delta(t) \longrightarrow v \quad (26)$$

so that

$$(s + Dvk^2 + v\Sigma_a) \tilde{\phi}(k, s) = v \quad (27)$$

$$\tilde{\phi}(k, s) = \frac{v}{s + Dvk^2 + v\Sigma_a} \quad (28)$$

a single simple pole at $s = -(Dvk^2 + v\Sigma_a)$, so with $\mathcal{L}^{-1}\{(s + \alpha)^{-1}\} = e^{-\alpha t}$,

$$\hat{\phi}(k, t) = v e^{-(Dvk^2 + v\Sigma_a)t} \quad (29)$$

and the Fourier inversion completes the square exactly as in Step 4:

$$\phi(x, t) = \frac{v}{2\pi} \int_{-\infty}^{\infty} e^{-(Dvk^2 + v\Sigma_a)t} e^{ikx} dk \quad (30)$$

$$= \frac{v e^{-v\Sigma_a t}}{2\pi} e^{-x^2/(4Dvt)} \int_{-\infty}^{\infty} e^{-Dvt(k - \frac{ix}{2Dvt})^2} dk \quad (31)$$

$$= \frac{v e^{-v\Sigma_a t}}{2\pi} e^{-x^2/(4Dvt)} \sqrt{\frac{\pi}{Dvt}} = \frac{v e^{-v\Sigma_a t}}{\sqrt{4\pi Dvt}} e^{-x^2/(4Dvt)} \quad (32)$$

which is (20), with $\Sigma_a = (1 - c)\Sigma_t$.

Parts 3(a) and 3(b): the comparison

Each figure fixes c and shows $t = 1, 2, 3, 4, 7, 15$. The vertical line marks the causal front $|x| = vt$, beyond which the exact solution is identically zero. Only $x \geq 0$ is plotted; all three solutions are even in x .

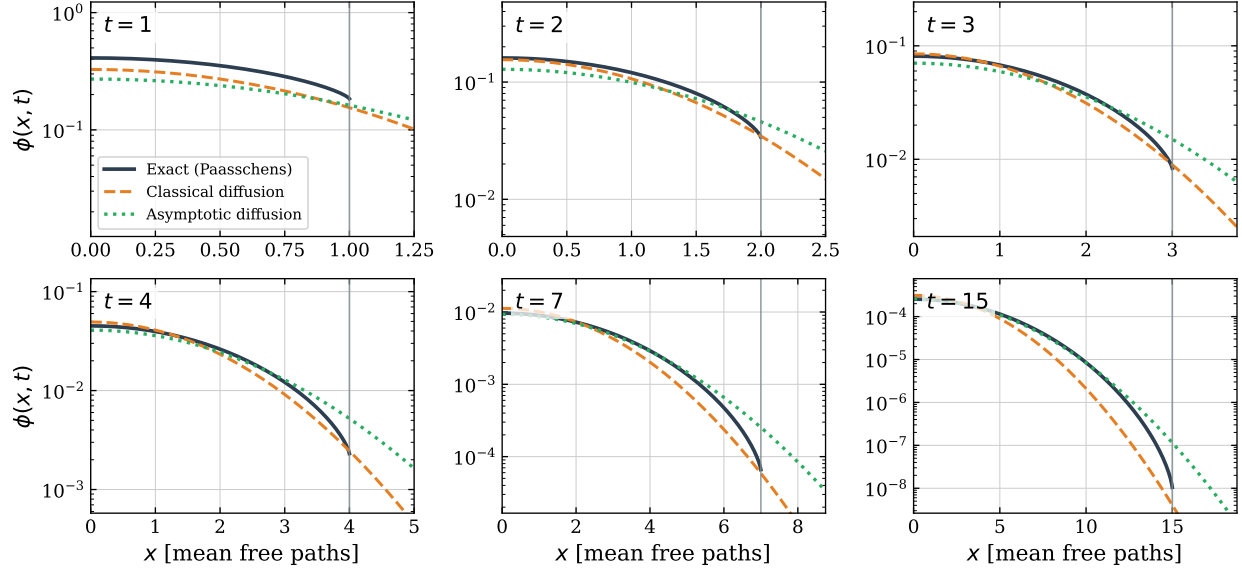


Figure 1: $c = 0.6$.

Part 3(c): the numerical solution

Before discretising, the absorption is removed exactly as in Step 2: with $\phi(x, t) = e^{-v\Sigma_a t}u(x, t)$ the term $v\Sigma_a\phi$ cancels identically and what is left is the pure heat equation carrying the pulse as its initial condition:

$$\frac{\partial u}{\partial t} = Dv \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0^+) = v \delta(x) \quad (33)$$

Since the equation and initial condition are even in x , so is the solution of (33).

Truncation. The solution is plotted in $[0, L]$ and is assumed to be zero at $x = L$. This truncation is justified because the spatial Gaussian from the analytic solution has variance $\sigma^2 = 2Dvt$, so taking

$$L = x_{\max} + 8\sqrt{2Dvt_{\max}} \quad (34)$$

places the truncation eight standard deviations beyond the last point to be plotted, where the solution is of order e^{-32} of its peak.

The grid, and the discrete pulse. Take nodes $x_j = jh$ with $j = 0 \dots N$. The spatial width is $h = L/N$, and time levels $t^n = n\Delta t$. On a node-centred grid the trapezoid weight of node 0 on the half line is $h/2$, and the full-line integral is twice the half-line one, so the cell straddling the origin carries total weight h :

$$\int_{-\infty}^{\infty} u(x, 0) dx \approx 2 \cdot \frac{h}{2} u_0^0 = h u_0^0 \stackrel{!}{=} v \quad \implies \quad u_0^0 = \frac{v}{h}, \quad u_{j \geq 1}^0 = 0 \quad (35)$$

so the spike of height v/h in the first cell is the correctly normalised delta — the factor of two from the symmetry and the half-weight of the boundary node cancel.

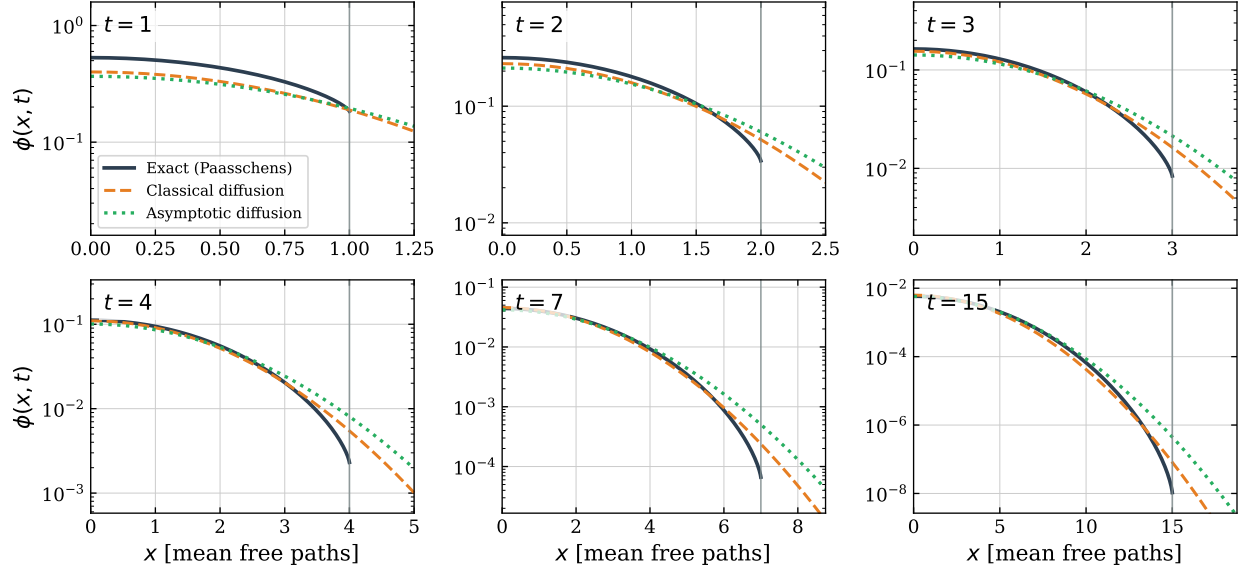


Figure 2: $c = 0.8$.

The update. Central second differences in space, evaluated at the old level, give the explicit scheme. With $r = Dv \Delta t / h^2$,

$$u_j^{n+1} = u_j^n + r(u_{j-1}^n - 2u_j^n + u_{j+1}^n), \quad 1 \leq j \leq N-1 \quad (36)$$

$$u_0^{n+1} = u_0^n + 2r(u_1^n - u_0^n) \quad (37)$$

$$u_N^{n+1} = 0 \quad (38)$$

the row at $j = 0$ following from the symmetry condition written as the ghost value $u_{-1} = u_1$.

Von Neuman Stability. Substituting the Fourier mode $u_j^n = \xi^n e^{ij\theta}$ into (36) gives the amplification factor

$$\xi(\theta) = 1 + r(e^{-i\theta} - 2 + e^{i\theta}) = 1 - 4r \sin^2(\theta/2) \quad (39)$$

and $|\xi| \leq 1$ for every θ requires $4r \leq 2$, i.e.

$$r \leq \frac{1}{2} \quad \Longleftrightarrow \quad \Delta t \leq \frac{h^2}{2Dv} \quad (40)$$

Conservation, as the check. Let M^n be the trapezoid mass on the half line, so that $\int_{-\infty}^{\infty} u dx \approx 2M^n$:

$$M^n = h \left[\frac{1}{2} u_0^n + \sum_{j=1}^{N-1} u_j^n + \frac{1}{2} u_N^n \right] \quad (41)$$

Calculating M^{n+1} using (36) and the boundary conditions gives

$$M^{n+1} - M^n = rh(u_N^n - u_{N-1}^n) = -rh u_{N-1}^n \quad (42)$$

So the scheme conserves particles to machine precision until the tail reaches the truncation, and the only leak is through the far boundary, at a rate that is directly observable as u_{N-1}^n . Restoring the prefactor, the check on the solver is $\int \phi dx = e^{-v\Sigma_a t}$, the same identity already used on the exact solution.

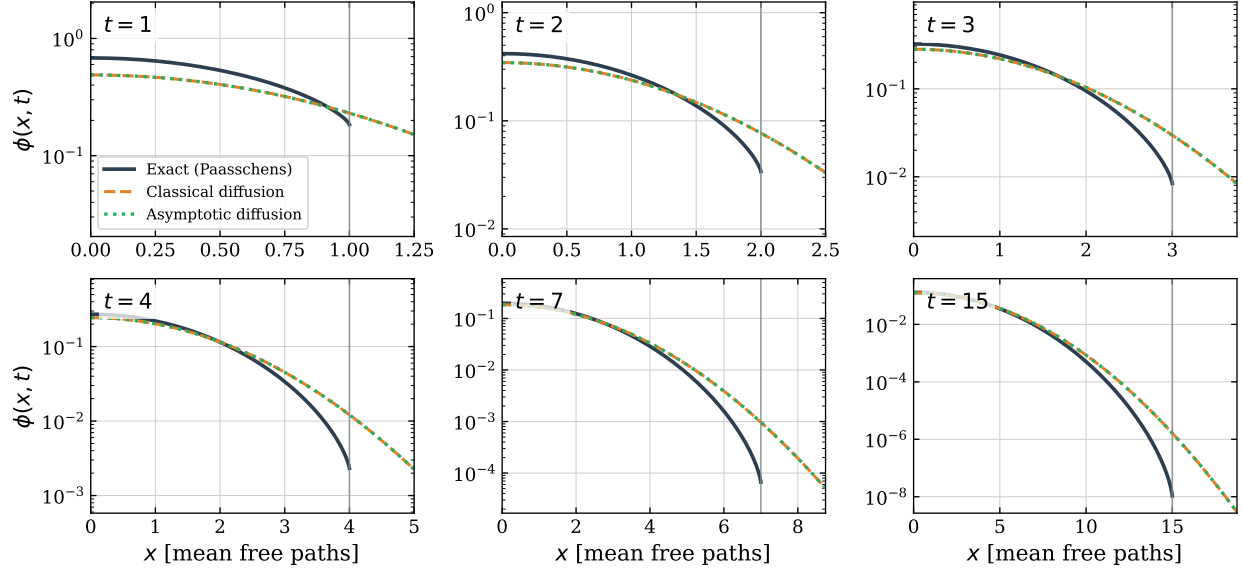


Figure 3: $c = 1$. The two diffusion coefficients coincide at $D_0 = 1/3$, so the classical and asymptotic curves overlies each other.

Results. The scheme is implemented in `src/homework2/solver.py` and run at $N = 2000$ nodes, which puts $h \approx 0.022$ and $\Delta t \approx 6.5 \times 10^{-4}$, i.e. about 2×10^4 sweeps to reach $t = 15$. Figure 6 places it over the closed form it discretises and Figure 7 measures what separates them. Three results are worth quoting:

- **Conservation is exact.** $\int \phi dx$ matches $e^{-(1-c)t}$ to between 0 and 1.1×10^{-15} over $c = 0.6, 1.0, 1.5$ and $t = 1, 4, 15$ — round-off, not discretisation, as (42) requires.
- **The scheme is second order.** The peak error at $t = 4$ falls $1.007 \times 10^{-3} \rightarrow 2.586 \times 10^{-4} \rightarrow 6.466 \times 10^{-5}$ as N doubles from 500 to 2000: ratios of 3.89 and 4.00.

Part 3(d): the numerical solutions against the exact one

Both marches of (36) — $D = 1/3$ and $D = D_0(c) = (1 - c)\nu_0^2$ — against the Question 2 flux carried to c by (1): pointwise in Figure 8, and, since both carry the same population $e^{-(1-c)t}$ so that $p = e^{(1-c)t}\phi$ are probability densities, as the single misplaced fraction of Table 1,

$$E = \left| \frac{\phi_{\text{diff}}}{\phi_{\text{exact}}} - 1 \right|, \quad \delta(t; c) = \frac{1}{2} \int |p_{\text{diff}} - p_{\text{exact}}| dx = \frac{1}{2} \left[\int_{|x| < vt} p_{\text{exact}} E dx + L \right] \quad (43)$$

that is, E averaged over the particles plus L , the flux put beyond the front. L dies: it is $\text{erfc} \sqrt{vt/4D}$, the front standing at $\sqrt{vt/2D}$ standard deviations, which grows as $\sqrt{t}$ because the packet widens as $\sqrt{t}$ while the front runs away linearly — 22.07%, 1.43%, 2.10×10^{-6} at $t = 1, 4, 15$, against the solver's 21.47%, 1.41%, 2.03×10^{-6} .

The rest does not. In density form (1) is $p(x, t; c) = cp(cx, ct; 1)$ and the $c = 1$ density relaxes onto the classical Gaussian of variance $2vt/3$, so the exact flux spreads with D_∞ — neither $1/3$ nor $D_0(c)$ — and two centred Gaussians of variances $2D_<vt$ and $2D_>vt$, crossing at $x^{*2} = 4vtD_<a^2 \ln a/(a^2 - 1)$,

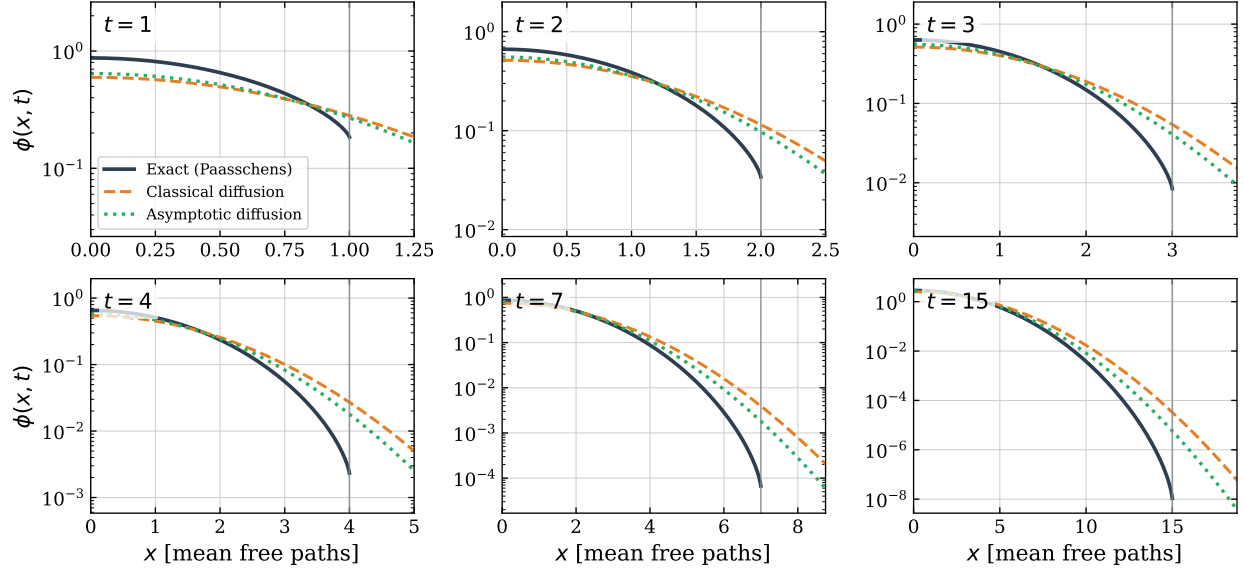


Figure 4: $c = 1.2$.

stay δ_∞ apart:

$$D_\infty(c) = \frac{1}{3c \Sigma_t}, \quad a^2 \equiv \frac{D_>}{D_<}, \quad \delta_\infty = \operatorname{erf} \sqrt{\frac{a^2 \ln a}{a^2 - 1}} - \operatorname{erf} \sqrt{\frac{\ln a}{a^2 - 1}} \quad (44)$$

free of t — a fixed variance ratio is a fixed distance however far both spread. So δ stops at $\delta_\infty(D, D_\infty(c))$, zero only at $c = 1$: by $t = 40$ the classical curves measure 11.48% against 12.29% at $c = 0.6$ and 10.12% against 9.78% at $c = 1.5$, while $c = 1$ is still falling. D_0 beats $1/3$ by a factor of three to nine at $t = 15$ and by more in the floors, but is not right either: $D_\infty = \frac{1}{3} + \frac{1}{3}(1 - c)$ against $D_0 = \frac{1}{3} + \frac{4}{15}(1 - c)$, four fifths of the first-order correction against none.

c	D	$\delta(t; c)$					δ_∞
		$t = 1$	$t = 2$	$t = 4$	$t = 7$	$t = 15$	
0.6	classical, $1/3$	22.09%	8.32%	5.59%	7.71%	10.12%	12.29%
	asymptotic, 0.48588	31.02%	15.70%	6.17%	2.12%	1.15%	3.24%
0.8	classical, $1/3$	22.09%	9.09%	2.18%	2.10%	3.73%	5.39%
	asymptotic, 0.39629	26.18%	12.94%	5.64%	2.60%	0.59%	1.21%
1.0	both, $1/3$	22.41%	11.20%	5.38%	3.00%	1.38%	0
1.2	classical, $1/3$	23.22%	13.63%	8.84%	6.89%	5.55%	4.41%
	asymptotic, 0.28717	19.62%	10.04%	5.24%	3.28%	1.94%	0.80%
1.5	classical, $1/3$	24.86%	17.07%	13.30%	11.76%	10.69%	9.78%
	asymptotic, 0.23745	16.66%	8.89%	5.11%	3.58%	2.51%	1.60%

Table 1: The misplaced fraction (43) and the floor (44) it settles on, from `check_total_variation` in `src/homework2/main.py`.

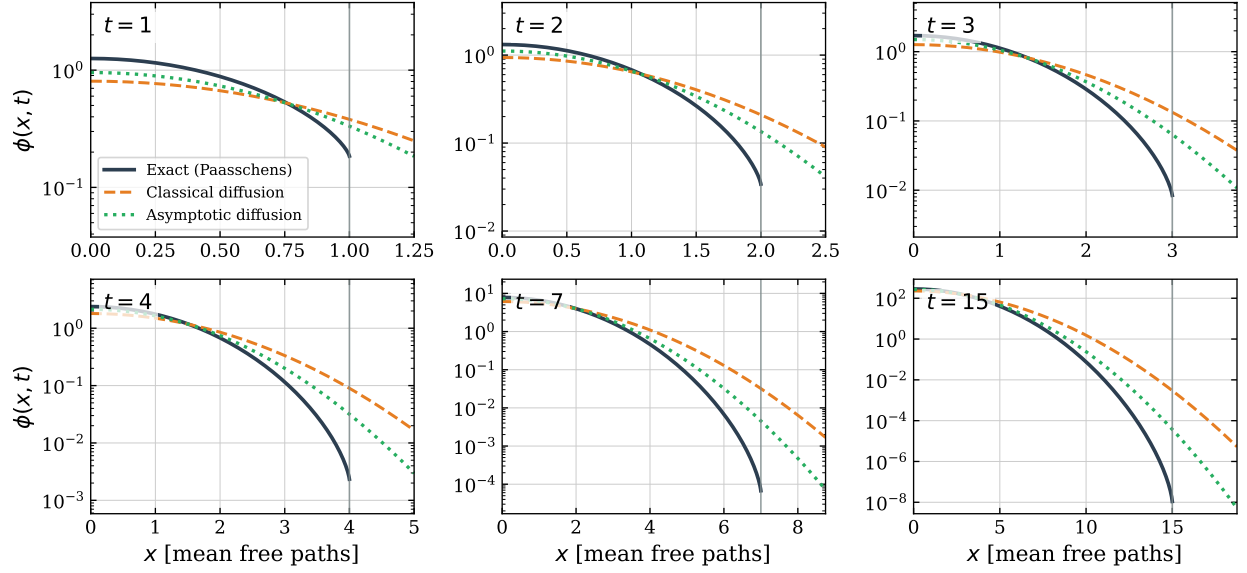


Figure 5: $c = 1.5$.

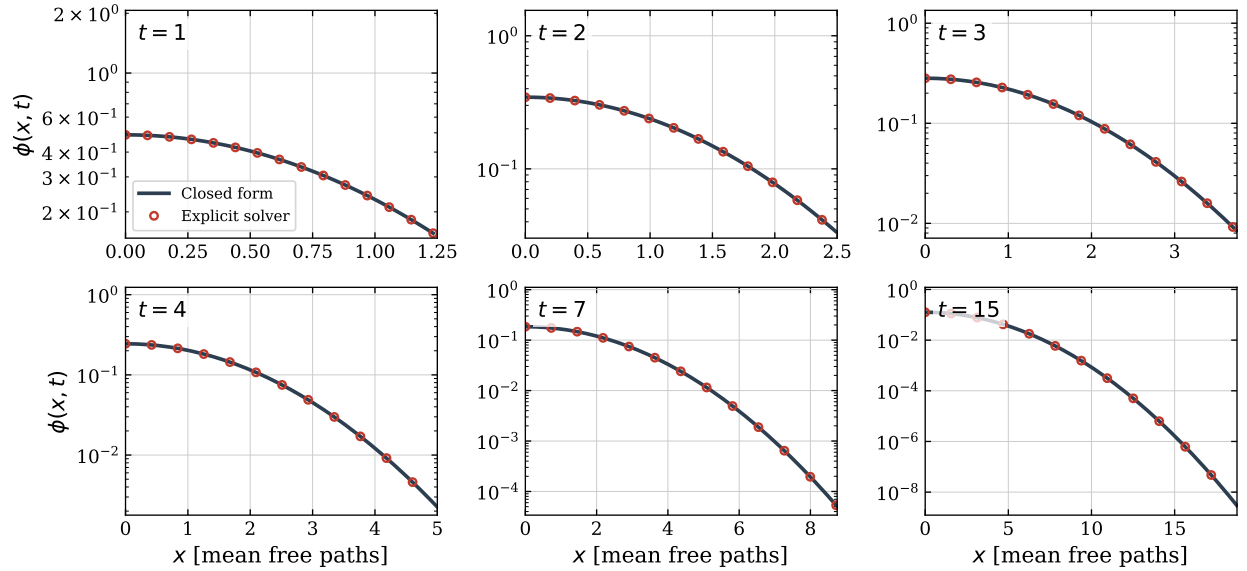


Figure 6: Part 3(c): the explicit solver (markers, thinned) over the closed-form diffusion Green's function it discretises (solid), at $c = 1$ and $N = 2000$. The two are indistinguishable at this scale over eight decades of flux; what separates them is Figure 7.

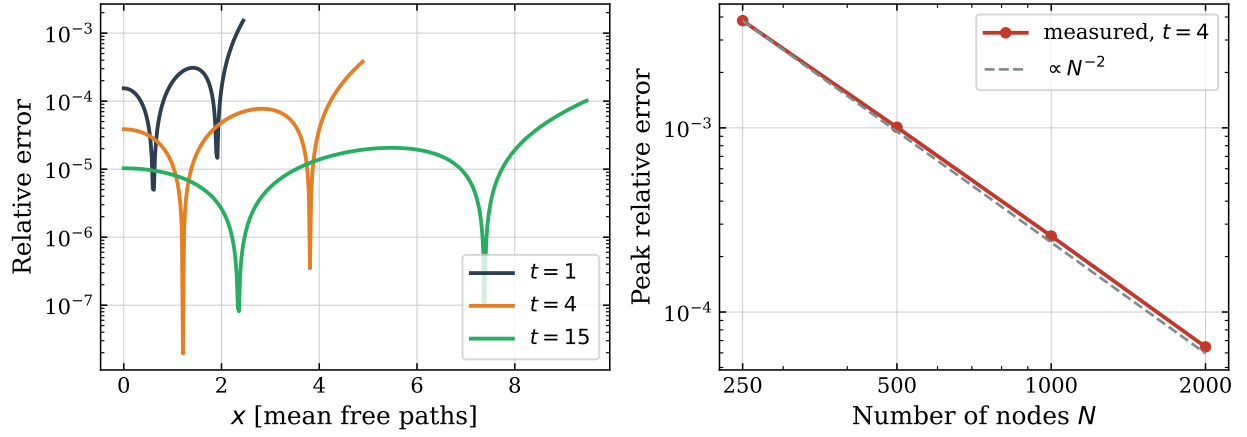


Figure 7: Part 3(c): the solver's relative error against the closed form, within three standard deviations. Left, across the domain at three times. Right, the peak error at $t = 4$ against the node count, against a second-order guide.

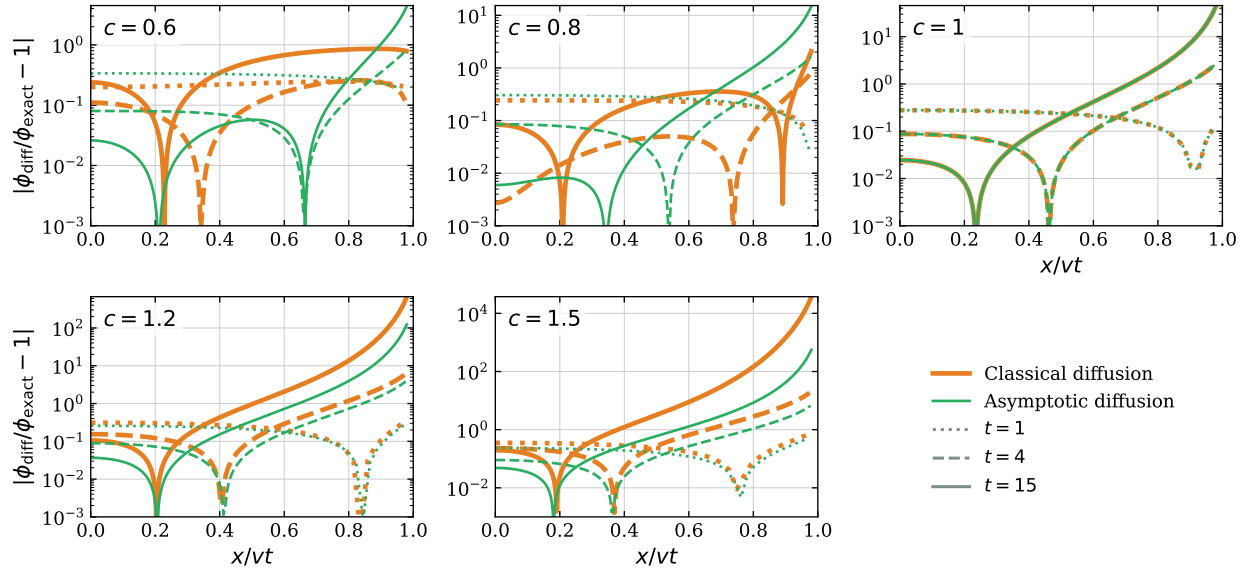


Figure 8: Part 3(d): the pointwise error E of (43), one panel per c , against the scaled position x/vt — cut at $0.98 vt$, since ϕ_{exact} stops at the front. The downward spikes are sign changes. Part 3(c) put the discretisation at 6.5×10^{-5} , so what is plotted is the approximation, not the solver. At $c = 1$ the two coefficients coincide.

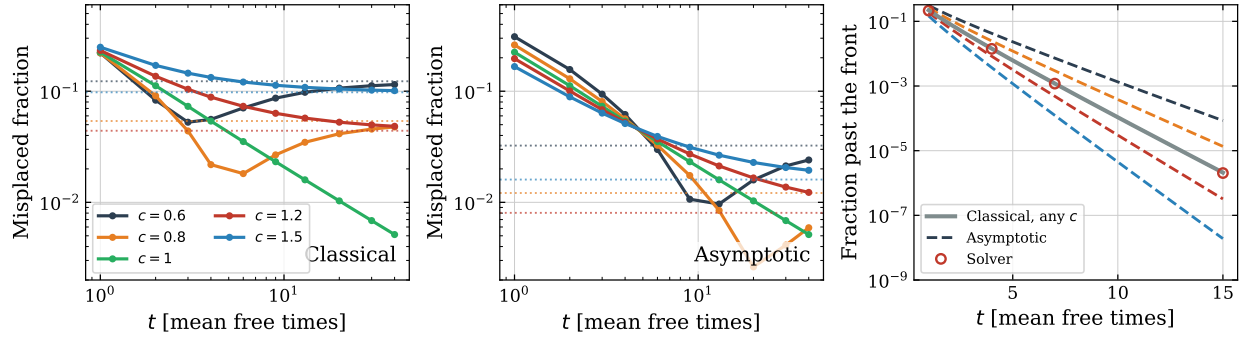


Figure 9: Part 3(d). Left and centre, on one scale: δ against time, with each c 's floor dotted beneath its curve. Every classical curve turns and climbs onto its floor; the asymptotic ones sit far lower, and the $c = 1$ curve, alone in having none, decays throughout. Right: the leak L of (43), which depends on c only through D — so the classical case is one curve for every c — with the solver's own leak marked on it.